

Biology and Health Sciences

I

002

30/07/2024

08:30 AM – 11:30 AM



ORDINARY LEVEL NATIONAL EXAMINATIONS, 2023-2024

SUBJECT: Biology and Health Sciences I

DURATION: 3 HOURS

INSTRUCTIONS:

- 1) Write your names and index number on the answer booklet as they appear on your registration form and **DO NOT** write your names and index number on additional answer sheets of paper if provided.
- 2) Do not open this paper until you are told to do so.
- 3) This paper consists of **three** sections **A**, **B** and **C**

Section A: Answer **ALL** questions. **(55 marks)**

Section B: Answer **THREE** questions. **(30 marks)**

SECTION C: This section is **compulsory**. **(15 marks)**

- 4) Use only a **blue** or **black** pen.

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SECTION A: Attempt all questions from this section.

(55 marks)

1. Which of the following blood cells contain haemoglobin? **(1 mark)**
 - a) Lymphocytes
 - b) Phagocytes
 - c) Red blood cells
 - d) Platelets
2. What is the function of the cornified layer of the epidermis of human skin? **(1 mark)**
 - a) Temperature regulation through sweating
 - b) Perception of sensation of temperature, pain and pressure
 - c) Synthesis of Vitamin D
 - d) Prevention of uncontrolled water loss by evaporation
3. Which acid is produced in human muscle when there is no enough oxygen? **(1 mark)**
 - a) Lactic acid
 - b) Hydrochloric acid
 - c) Citric acid
 - d) Ascorbic acid
4. Which of these statements below is not biologically correct?
 - a) All mammals are vertebrates
 - b) Frog's body is moist or rough skin
 - c) Birds have feathers
 - d) Amphibians have scales
5. Which of these is not a function of joints?
 - a) Articulations
 - b) Allows bone growth
 - c) Enables mobility
 - d) Supports body
6. Complete the following table with the corresponding type of contraceptive method: **hormonal or non-hormonal or barrier.****(5 marks)**

Method	Type
Spermicide	
Condom	
Implant	
IUD	
Pills	

7. State whether each of the following statements is **true** or **false**. (6 marks)

- a) Mutations cannot be caused by copying errors in the genetic material during cell division.
- b) Mutations can be caused by environmental changes.
- c) Mutations can lead to death
- d) Albinos are results of mutation in humans
- e) Genotypic variations are visible
- f) Phenotypic variations are visible

8. Fill the blank spaces in the paragraph below choosing word from the following list: **meiosis, fertilization, parent cells, haploid, diploid, gametes**. One word can be used more than one time. Write the full paragraph and underline the added word. (7 marks)

The process of meiosis produces unique reproductive cells called which have half the number of chromosomes as the parent cell. During the fusion of gametes from two individuals, restores the diploid condition. Gametes are (containing only one set of chromosomes) while the zygote is (containing two sets of chromosomes). Thus, sexually reproducing organisms alternate between haploid and diploid stages. In sexual reproduction, two individuals produce offsprings that have characteristics from both parents.
The gametes are produced from via the process called

9. Match the following blood cells with their functions: (3 marks)

Blood component	function
White blood cells	Defend the body against pathogens
Red blood cells	Clotting of blood
platelets	transport of respiratory gases

10. Write **Yes** if the statement is correct and **No** if it is wrong. (3 marks)
- a) ATP is released in the mitochondrion.
 - b) Aerobic respiration releases more energy than anaerobic respiration.
 - c) Cellular respiration occurs only in animal cells.
11. Identify the parts of human reproductive system in which each of the following gametes is produced. (2 marks)
- a) Sperm cell
 - b) Ovum

12. The age of four individuals is as follows:

Individual	Age
A	75-year-old grandfather
B	35-year-old father
C	28-year-old pregnant mother
D	D: 3-year-old boy

- a) Between the boy and the grandfather, who needs more milk? **(1 mark)**
 - b) Give reasons for your answer in (a). **(2 marks)**
 - c) Explain why a 28 year old pregnant mother needs more iron than a man of the same age. **(2 marks)**
13. Athletes are normally given glucose and not sucrose. Give a reason for this. **(3 marks)**
14. What is the basic measure can you take to protect yourself from contracting a disease? **(2 marks)**
15. Explain why farmers who grow cereals such as wheat, barley and rice must ensure that plants grow close to each other. **(2 marks)**
16. In your opinion, how would you deal with sexual assault? **(2 marks)**
17. Why should industrial plants treat their waste water before dumping in water bodies? **(4 marks)**
18. Fill in the blank spaces. **(3 marks)**
- a) Antigens bind at thesite formingcomplexes.
 - b) immunity is a naturally acquired immunity.

SECTION B: Attempt only three questions (30 marks)

19. a) How can we help someone with cholera? **(2 marks)**
b) List any four practices to ensure good health. **(8 marks)**
20. Describe the negative impact of monoculture system to the biological diversity in an ecosystem. **(10 marks)**
21. Answer the following in relation to blood.
- a) What is blood plasma? **(1 mark)**
 - b) Give the role of blood plasma. **(2 marks)**
 - c) The ABO blood group system has four blood groups. Name them. **(4 marks)**

- d) Blood groups can be positive (+) or negative (-) according to whether the blood contains or does not contain a particular factor. Name this factor. **(1 mark)**
- e) Suggest a reason why it is important to know a person's blood group. **(1 mark)**
- f) What is the function of platelets in the blood? **(1 mark)**
22. A man whose blood group is A marries a woman whose blood group B. If one of their children is blood group O:
- a) What are the genotypes of their parents? **(2 marks)**
- b) Use a Punnett square to determine the genotype of the other children. **(8 marks)**
23. a) What do you understand by the term mitosis? **(2 marks)**
- b) "Mitosis is essential in all living organisms". Give reasons to support your answer. **(8 marks)**

SECTION C: THIS QUESTION IS COMPULSORY (15 marks)

24. Assume you are performing an experiment to demonstrate osmosis.
- a) Enumerate the apparatus and chemicals you may use in your demonstration. **(5 marks)**
- b) Suggest a control that you might use in this activity. **(2 marks)**
- c) State the result(s) of the demonstration. **(2 marks)**
- d) Briefly explain the result(s). **(2 marks)**
- e) Suggest how storing food in salty water prevents the growth of bacteria or fungi on the stored food. **(2 marks)**
- f) Apart from the method described in (e) above, suggest any other way of preventing food spoilage. **(1 mark)**

-END-

**MARKING SCHEME OF ORDINARY LEVEL BIOLOGY
NATIONAL EXAMINATION 2023-2024**

SECTION A: Answer all questions

- 1) (c) Red blood cells
- 2) (a) Temperature regulation through sweating
- 3) (a) Lactic acid
- 4) (d) Amphibians have scales
- 5) (a) Articulations accept also (b)

6)

Method	Type
Spermicide	Barrier accept also <i>Non-Hormonal</i>
Condom	Barrier
Implant	Hormonal
IUD	Non-hormonal also accept hormonal or barrier
Pills	Hormonal

7)

- a) False
 - b) True
 - c) True
 - d) True
 - e) False
 - f) True
- 8) The process of meiosis produces unique reproductive cells called **gametes** which have half the number of chromosomes as the parent cell. During **fertilization** the fusion of gametes from two individuals, restores the **diploid** condition. Gametes are **haploid** (containing only one set of chromosomes) while the zygote is **diploid** (containing two sets of chromosomes). Thus, sexually reproducing organisms alternate between haploid and diploid stages. In sexual reproduction, two individuals produce **diploid** offsprings that have characteristics from both parents. The gametes are produced from **parent cells** via the process called **meiosis**.

9)

Blood component	function
White blood cells	Defend the body against pathogens
Red blood cells	Clotting of blood
platelets	transport of respiratory gases

10) a) Yes

b) Yes

c) No

11) a) Testis / testicle / male gonads

b) Ovary / Ovaries / female gonads

12)

a) D: 3-year-old boy.

b) Milk contains important proteins which are very essential for the **growth/Body building /development** and repair of the body.

Children have growing bodies than adults. Hence they are given more milk than grandfather who is adult.

c) A 28 year old mother needs extra iron because:

- Formation of more blood in both foetus and mother.
- Formation of haemoglobin
- Formation of red blood cells
- Avoid anaemia
- Transport of more oxygen

13) **Glucose is a simple sugar/monosaccharide** which can be broken down easily to produce energy.

The Absorption of glucose is also very fast in the blood.

- No need of its digestion

14) Protection/ keeping proper hygiene

Keeping personal hygiene by washing your hands often with soap and clean water

Using sanitizer, cleaning and disinfecting surfaces that are often touched

Cover the food in cleaned utensils and avoid drinking non boiled water.

15) To allow cross pollination to take place easily.

To ensure high yield production

To ensure support each other against wind

To allow maximum absorption of water/ Easy irrigation

To maximize land use

- 16) Report to authorities, parents, teachers and guardians.
 Help the victim to avoid to be isolated
 Giving the victim medical assistance in case sexual assault is attempted/to take the victim to the ISANGE ONE STOP CENTER / Hospital / Health center.
 Avoiding peer pressure
 Counseling
 Avoid drugs abuse
 Education/ sensitization
 Consider legal actions
 Say no
 Gathering and keeping physical evidences
- 17) Industrial effluent should be treated for safe use on the land that is used as manure or for disposal.
 Industrial plants treat their waste water before dumping in water bodies to prevent pollution.
 For safe use on the land that is used as manure or for disposal
 To prevent pollution
 To prevent eutrophication
 Avoid acidification of water bodies
 Prevent water-borne diseases
 Production of biogas
 Avoid death of aquatic organisms
 For being reused.
- 18)
- a) Antigens bind at the **active /antibody/ antigen-binding/ binding** site forming **antigen-antibody** complexes.
 - b) **Passive / active / adaptive** immunity is a naturally acquired immunity.

SECTION B: Attempt only three questions

- 19) a) Take the patient immediately to the hospital for treatment. Give the person rehydration salts/fluids
 Give the patient the Intravenous fluids and electrolyte.
- b) Daily practices to ensure good health include:
- i) Brushing our teeth three times in a day and every after a meal.
 - ii) Practicing moderate physical exercises
 - iii) Healthy eating: balanced diet, avoid too much fats/eating at close intervals
 - iv) Vaccination and immunisation practices
 - v) Regular medical check-up in hospitals
 - vi) Sanitation i.e keeping our surroundings clean.

- vii) Enough sleep
- viii) Having protected sexual intercourse/ abstinence
- ix) Drinking enough clean water and stay hydrated
- x) Reduce sugary food and limit sugared beverages
- xi) Avoid excessive alcohol consumption and drug abuse

20) Monoculture crops have no genetic variability because they are not allowed to reproduce naturally. This exposes their pests and weeds to environmental adaptations and become resistant to chemicals used to control them as the only way to control pests in this set up is to use more chemical control mechanisms

Susceptibility to pests: The pests and weeds that attack them adapt easily to the environment and develop resistance to chemicals used to control them.

Environment pollution through the use of dangerous synthetic chemicals/ Eutrophication

May cause eutrophication that affect aquatic life

Use of fuel energy in combination with the chemical fertilizers and pesticides, cause environment pollution (Air, soil and water) and lead to climate change

May cause habitat destruction through searching enough space for growing crops for rising demand for food and other agricultural products

Extinction of some species

Soil erosion

Use of dangerous synthetic chemicals

It leads to soil degradation

Drought

Extreme weather condition

Use of biofuels

Conflict

Food wastage

21)

- a) Liquid part of blood
- b) Plasma transports blood cells, nutrients such as glucose and amino acids; transports antibodies, metabolic waste products...
 - Transportation/distribution of blood cells
 - Transportation /distribution of antibodies
 - Transportation / distribution of waste substances such as carbon dioxide and urea
 - Transportation / distribution of Hormones such as adrenaline, insulin and glucagon
- c) White blood cell: Protects the body against microorganisms.
- d) A or B or AB or O
- e) +/- factor: Rhesus factor or Rh factor
- f) For proper blood transfusion, **to avoid incompatibility of blood/agglutination/** in case of blood transfusion / **For Safe organ transplant.**

g) Platelets are responsible of blood clotting.

22) a) Genotypes of the parents: AO and BO or $I^A I^O$ and $I^B I^O$

b) Phenotype of parents: group A x group B

Genotype of parents: AO x BO

Gametes: A, O and B, O

Punnet square

Father Mother	A	O
B	AB	BO
O	AO	OO

The genotypes of their children are:

25% or $\frac{1}{4}$ AB

25% or $\frac{1}{4}$ BO

25% or $\frac{1}{4}$ AO

25% or $\frac{1}{4}$ OO

Or

Phenotype of parents: group A x group B

Genotype of parents: $I^A I^O$ x $I^B I^O$

Gametes: I^A, I^O and I^B, I^O

Punnet square

Father Mother	I^A	I^O
I^B	$I^A I^B$	$I^B I^O$
I^O	$I^A I^O$	$I^O I^O$

The genotypes of their children are:

25% or $\frac{1}{4}$ $I^A I^B$

25% or $\frac{1}{4}$ $I^B I^O$

25% or $\frac{1}{4}$ $I^A I^O$

25% or $\frac{1}{4}$ 10¹⁰

23) a) Mitosis: is a type of cell division in which a parent cell divides itself into two identical daughter cells which are genetically the same as the parent cell.

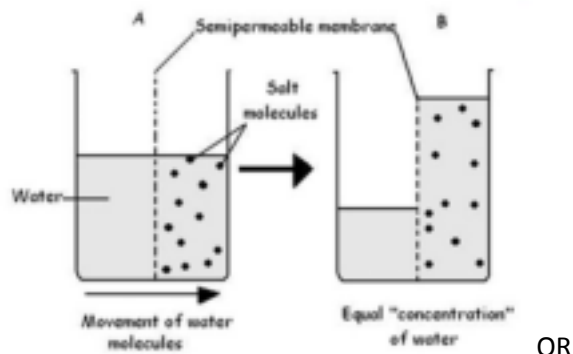
b) Reasons:

- ✓ It facilitates repair of worn out body tissues
- ✓ It leads to body building hence growth
- ✓ It brings about regeneration of parts of organisms e.g. in reptiles
- ✓ It is responsible of replacement of worn out cells
- ✓ It provides a basis for asexual reproduction
- ✓ It maintains genetic stability/the number of chromosomes in the nucleus is retained.

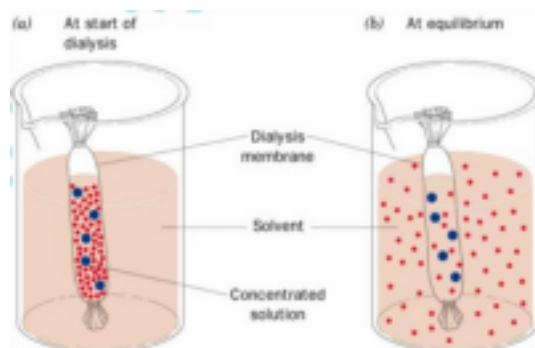
SECTION C: THIS QUESTION IS COMPULSORY

24)

a) Diagram of osmosis set-up: Two solutions and a membrane
Labels: e.g. water; salt or sugar solution; visking tubing or potato; beaker



OR



- b) Control: e.g. Only water in visking tubing and in the beaker; only water around the semi permeable membrane.
- c) Result: The visking tubing increase in size if it contains a salt or sugar solution. Or water moves from hypotonic solution to hypertonic

